

a Road junctions are planar-lattice coordinations

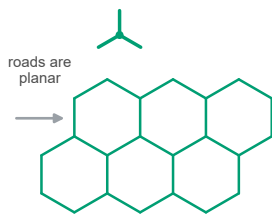
CEBH $p_c \approx 0.47$

overstates robustness



Random graph
(locally tree-like)

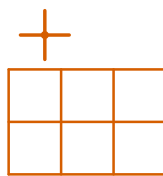
$p_c = 0.653$



roads are
planar

3-way junction
= honeycomb

$p_c = 0.500$



4-way junction
= square



planar embedding
(no free crossings)

Real road network

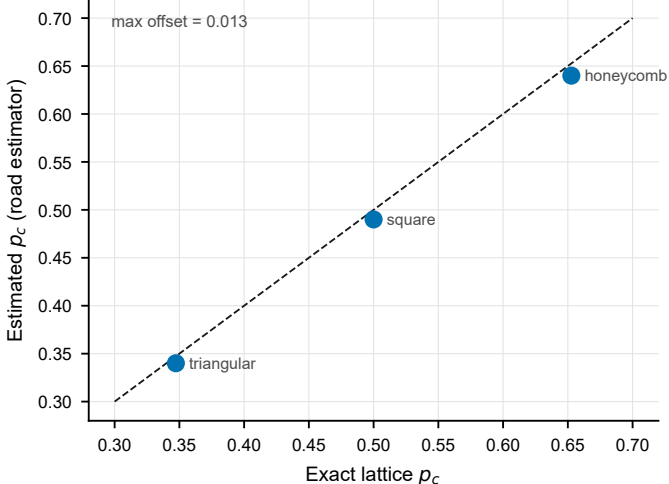
$$p_c = w_3 \cdot 0.653 + w_4 \cdot 0.500$$

w_3, w_4 : 3-way and 4-way shares

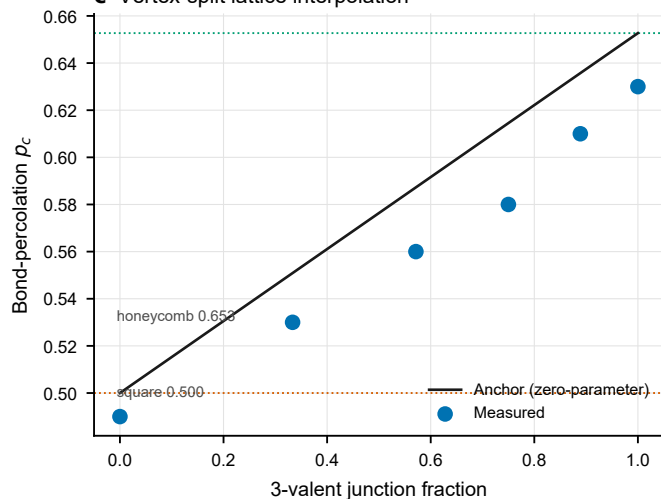
observed $p_c \approx 0.62-0.69$

predicted with no fitting

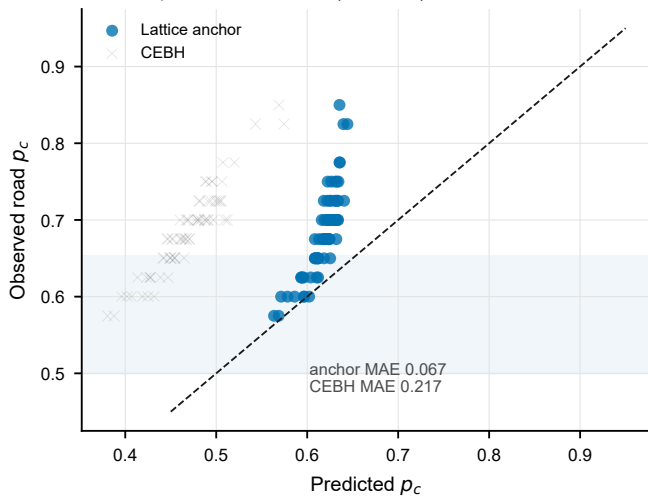
b Estimator reproduces lattice thresholds



c Vertex-split lattice interpolation



d Zero-parameter anchor (71 cities)



e Error: degree-moment vs lattice

